

AI-Powered Supply Chain Conversational Knowledge Assistant for Small and Medium Enterprises

The proposed project aims to develop an AI-powered conversational knowledge assistant designed specifically for Small and Medium Enterprises (SMEs) to improve access to supply chain knowledge and operational information. Many SMEs face challenges in managing organizational knowledge due to the absence of dedicated supply chain teams, advanced enterprise software, and centralized information systems. Important information is often scattered across documents, spreadsheets, emails, manuals, and reports, making it difficult for employees and managers to quickly obtain the information they need.

The objective of the proposed system is to provide an easily accessible conversational platform through which users can interact with company knowledge using natural language. Instead of manually searching through multiple documents or depending on experienced employees for information, users can simply ask questions and receive accurate responses instantly.

The system will utilize the concept of Retrieval-Augmented Generation to retrieve relevant information from the company's knowledge repository and generate responses based only on verified sources. This ensures that answers are accurate, transparent, and supported by organizational documents rather than relying on assumptions or generated content.

The knowledge repository will consist of business and supply chain documents commonly used by SMEs, including:

- Procurement policies and supplier information
- Inventory management procedures
- Warehouse operating guidelines
- Transportation and logistics procedures
- Purchase records and supplier agreements
- Product catalogs and specifications
- Company standard operating procedures
- Training manuals and work instructions
- Quality control documents
- Regulatory and compliance documents

Users will interact with the system through a conversational interface and ask questions in natural language without requiring technical knowledge or predefined commands.

Examples of questions include:

- What is our reorder policy for packaging materials?
- Which suppliers provide raw materials for Product A?
- What is the standard receiving procedure for incoming inventory?
- What is the lead time for imported materials?
- What is the process for handling customer returns?
- What are the storage requirements for finished products?
- Which products are currently low in stock?

For each query, the system retrieves relevant information from the knowledge base and generates responses with supporting references such as source documents and relevant sections. This improves trust and allows users to verify information whenever necessary.

To complement the conversational assistant, the project will include an operational dashboard designed specifically for SME owners and managers to provide a simple overview of business and supply chain activities.

The dashboard will provide information such as:

- Current inventory status
- Products approaching reorder levels
- Supplier information and lead times
- Pending purchase orders
- Recent stock movements
- Product availability status
- Monthly purchasing trends
- Sales and demand summaries
- Warehouse stock distribution
- Supplier performance indicators

The dashboard is designed to provide quick visibility into business operations without requiring complex enterprise software solutions.

The system is intended for use by SME owners, procurement personnel, warehouse staff, supervisors, and employees who require fast access to operational knowledge and supply chain information.

The proposed solution aims to reduce information search time, improve operational efficiency, support employee training, preserve organizational knowledge, and enable SMEs to benefit from modern artificial intelligence technologies without significant investment in large-scale enterprise systems.

The final output of the project will be a web-based AI-powered conversational knowledge assistant integrated with an easy-to-use operational dashboard that enables SMEs to access organizational knowledge and supply chain information through natural language interaction.